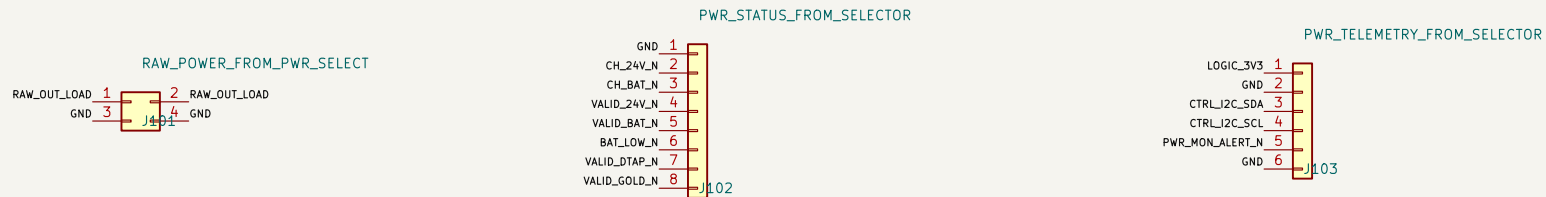


A. PWR-SELECT INTERFACE



Mates with PWR-SELECT J301; two contacts per polarity validate 15 A derating.

Mates with PWR-SELECT J401; add 3.3 V pull-ups on the carrier.

Mates with PWR-SELECT J402; carrier supplies 3.3 V and owns I2C pull-ups.

B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL

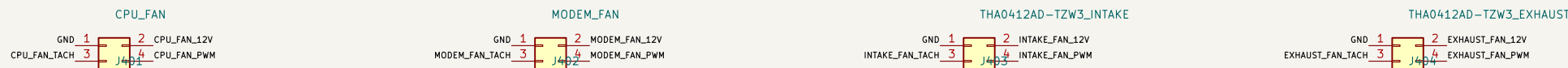


Separate filtered audio-power harness; no fan/radio loads.

Mates with AUDIO-BXB J101 using Molex 87832-6423 headers; harness assembly remains vibration-qualified.

Carrier drives MCLK/BCLK/FSYNC/DAC data; AUDIO-BXB drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS



J403 is physically keyed/labeled INTAKE; J404 is keyed/labeled EXHAUST.

D. DETAILED CAPTURE SUITE – REV A1

01 CMS-Core-Allocated; exact 300-contact CMS symbol, 76 locked functions

02 CMS-Core-Allocated; VCC, SYSIN, reset, recovery, power-on and debug UART

03 Network-PCIe: PITC9X256086P, 3x LAN7430 and native WAN1 front end

04 Network-PCIe: Nx Wurth MagJack and Molex 0679101002 Mini PCIe 4TAR Wi-Fi

05 WWAN-SIM: TE 2199230-3 B-key, USB3/USB2, controls and dual-SIM mux

06 Display-Harness: HDMI, USB touch and dedicated 12 V / 2.5 A branch

07 Audio-Control: I2SO TDM, I2S1 ES8316 headset and CTIA amplifier path

08 Thermal-ID: I2C translation, GPIO expansion, sensors and four fan channels

09 Power-Regulators-A1: protected raw hold-up, all main rails, sequencing and test points

10 Audio-Bxb: all XLR shields bonded to chassis with one controlled AGND bond

11 Review: every detailed sheet has zero ERC errors; warnings are classified

DC/DC compensation and magnetics remain a separate power-design calculation milestone.

High-speed routing starts only after stackup and connector 2 datums are released.

The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.